

School of Electrical Engineering and Computer Science,
Washington State University,
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RESEARCH SUMMARY

My general research interests are in **Artificial Intelligence (AI)** and **Machine Learning (ML)**, with a main focus on developing robust, safe, and trustworthy ML algorithms with theoretical guarantees for deployment in critical real-world challenges within safety-sensitive domains (e.g., healthcare). My current work focuses on:

- Building trustworthy Large Language Model (LLM) hallucination control policies.
- Advancing conformal prediction-based uncertainty quantification with theoretical guarantees to facilitate effective human–ML collaboration (e.g., Clinician-in-the-Loop), including deployed prototypes for UTI detection and beyond.
- Developing uncertainty-aware energy management methods for wearable Internet of Things (IoT) devices for mobile health applications and beyond.

EDUCATION

Ph.D., Computer Science Jul., 2022 – May, 2027
Washington State University
Advisors: [Prof. Jana Doppa](#)

Thesis: Developing Novel Algorithms and Theory for Robust and Trustworthy Machine Learning.

M.Sc., Computer Science 🎓 Aug., 2022 – May, 2025
Washington State University
Thesis: *Conformalized Uncertainty Regions for Machine Learning-Based Multiple Cognitive Health Measures from Smartwatch Sensor Data.*

B.Sc., Statistics 🎓 (First Class Honors) Sept., 2014 – Nov., 2018
University of Nigeria
Thesis: *Sensitivity of Randomization test and F-test in Some Experimental Designs*
Class rank: *Best Graduating Student*

PROFESSIONAL APPOINTMENTS

Research Assistant Aug 2022 – Present
EECS Department – Washington State University
– Developing Novel Algorithms and Theory for Robust and Trustworthy Machine Learning.

Teaching Assistant and Guest Lecturer Aug 2022 – Dec 2024
EECS Department – Washington State University
– CptS 223: Advanced Data Structures C/C++ (Fall-2022)
– CptS 315: Introduction to Data Mining (Spring-2023)
– CptS 223: Advanced Data Structures C/C++ (Fall-2023)
– CptS 315: Introduction to Data Mining (Spring-2024)
– CptS 223: Advanced Data Structures C/C++ (Fall-2024)

Assistant Lecturer 2019 – 2020
Statistics Department – Kano State Polytechnic
– Course: Introduction to Statistical Inference

AWARDS AND HONORS

🏆 Mahmoud M. Dillsi Graduate Fellowship	Oct., 2025
🏆 Nakahara Tsuyoshi and Mary Fellowship	Nov., 2024
🏆 Outstanding Graduate Teaching Assistant in EECS Award <i>Voiland College of Engineering, Washington State University</i>	Apr., 2024
🏆 EducationUSA Opportunity Fund Program (OFP) Scholar. <i>USA Embassy, Abuja, Nigeria</i>	Jun., 2022
🏆 Best Graduating Student <i>Statistics Department – University of Nigeria</i>	Nov., 2018
🏆 Winner: Nigerian Statistical Association (NSA) competition <i>Representative of the University of Nigeria.</i>	May, 2018
🏆 Gold medalist: NSA competition individual category <i>Representative of the University of Nigeria.</i>	May, 2018

PUBLICATIONS

1. [AAAI'26] Chibuike E. Ugwu, Roschelle Fritz, Diane J. Cook, and Janardhan Doppa. **Clinician-in-the-Loop Smart Home System to Detect Urinary Tract Infection Flare-Ups via Uncertainty-Aware Decision Support.** *40th AAAI Conference on Artificial Intelligence (AAAI)* 2026.
Acceptance rate: 17.6%
2. [IJCAI'25] Chibuike E. Ugwu*, Dina Hussein*, Ganapati Bhat, and Janardhan Doppa. **Sustainable Wearables for Health Applications and Beyond via Uncertainty-Aware Energy Management.** *Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI)*, 2025 (* denotes equal contribution)
Acceptance rate: 18.5%
3. [DAC'25] Chibuike E. Ugwu*, Dina Hussein*, Ganapati Bhat, and Janardhan Doppa. **Uncertainty-Aware Energy Management for Wearable IoT Devices with Conformal Prediction.** *Proceedings of the 62nd ACM/IEEE Design Automation Conference (DAC)*, 2025. (* denotes equal contribution)
Acceptance rate: 22.5%
4. [ACM TECS'25] Pratyush Dhingra, Chibuike E. Ugwu, Janardhan Rao Doppa, and Partha Pratim Pande. **ERGo: Energy-Efficient Hybrid Graph Neural Network Training on Heterogeneous Processing-In-Memory Architecture.** *ACM Transactions on Embedded Computing Systems (TECS)*, 2025.
5. [QREI'23] Abimibola V. Oladugba, Chibuike E. Ugwu, and Uchenna C. Onwuamaeze. **Sensitivity and robustness of randomization test and F-test in some experimental designs.** *Quality and Reliability Engineering International*, 39(7), 2967-2974, 2023.

SOFTWARE PROTOTYPES

My goal is to democratize uncertainty quantification (UQ) in real-world, safety-sensitive applications by building practical tools based on calibration methods such as conformal prediction, which provide formal coverage guarantees. To this end, I have developed APR, URIC, and CCI.

Adaptive Prediction Region (APR) [\[code\]](#) Aug., 2024
– Developed a novel framework (APR) for constructing trustworthy, adaptive uncertainty regions for multi-target regression tasks.

Uncertainty Regions via Importance-weighted Calibration (URIC) [\[code\]](#) Nov., 2025
 – Designed and implemented software to produce calibrated predictions of multiple clinical cognitive health measures from continuous smartwatch sensor data.

Clinician-in-the-Loop Conformal-Calibrated Intervals (CCI) [\[code\]](#) Jan., 2026
 – Built a clinician-facing prototype that surfaces calibrated uncertainty for UTI risk detection and supports human feedback in the decision loop.

CONFERENCE PRESENTATIONS

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| 1. AAAI Conference on Artificial Intelligence (AAAI) | 2026 |
| 2. Annual Conference on Neural Information Processing Systems (NeurIPS) | 2025 |
| 3. International Joint Conference on Artificial Intelligence (IJCAI) | 2025 |

PROFESSIONAL SERVICES AND OUTREACH ACTIVITIES

PROGRAM COMMITTEE MEMBER

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| 1. International Conference on Learning Representations (ICLR) | 2026 |
| 2. Association for the Advancement of Artificial Intelligence (AAAI) | 2026 |
| 3. AAAI Conference on Artificial Intelligence, AI for Social Impact Track (AISI) | 2026 |
| 4. AAAI Conference on Artificial Intelligence, AI for Innovative Applications (IAAI) | 2026 |
| 5. International Conference of Machine Learning (ICML) | 2025 |
| 6. Association for the Advancement of Artificial Intelligence (AAAI) | 2025 |
| 7. AAAI Conference on Artificial Intelligence, AI for Social Impact Track (AISI) | 2025 |
| 8. AAAI Conference on Artificial Intelligence, AI for Social Impact Track (AISI) | 2024 |

VOLUNTEER AND OUTREACH ACTIVITIES

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|---|------|
| 1. Volunteer for AAAI Conference on Artificial Intelligence (AAAI), | 2026 |
| 2. Volunteer for International Joint Conference on Artificial Intelligence (IJCAI), | 2025 |
| 3. Mentor and Judge for Digital AgAthon (AgAID Institute), | 2025 |
| 4. Instructor for WSU Summer Programming Camp for Middle Schoolers, | 2025 |